

DAP I/V Stage Rev7

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Attribution

- **Author:** Alexander Loker, responsible for design decisions, component selection, analysis, and verification
- **Drafter/documenter:** Claude, writing from Alexander's direction and source data

Part Identification

Four I/V (current-to-voltage) channels convert the ES9038Q2M's differential current outputs (DACL, DACLB, DACR, DACRB) to voltages.

Signal chain per channel: I/V ($\times 2$) $\rightarrow$ differential summing + MFB low-pass $\rightarrow$ OPA1622 $\rightarrow$ headphone jack.

Op-amp: OPA1612AIDRGR (TI, dual, 8-SON 3×3 mm). Two packages cover the four I/V channels.

Design Requirements

- **DAC output mode:** current mode. ESS specifies its headline figures (128 dB DNR, -120 dB THD+N) in differential current mode (p. 52).
- **Maximum headphone output:** 0.6 V RMS (0.85 V peak).
- **Gain structure:** run the I/V stage as hot as the rails allow, and attenuate toward 0.6 V RMS later in the chain.
- **Op-amp distortion:** below the DAC's so the DAC stays the limit.

Design Constraints

- ± 3.3 V supply rails (6.6 V total)
- Low noise
- Small footprint

DAC Output Data (ES9038Q2M, p. 52)

Parameter	Value
Output impedance R_DAC	774 Ω $\pm$ 11%
Voltage-mode range V_OPP	$0.906 \times AVCC = 2.99 \text{ V p-p}$ at $AVCC = 3.3 \text{ V}$
Voltage-mode offset V_OCM	$AVCC/2 = 1.65 \text{ V}$
Full-scale current per pin	$V_OPP / R_DAC = 3.863 \text{ mA p-p}$
Current-mode offset	$(V_OCM - V_G) / R_DAC$

I/V Operating Point

$V_G = 0 \text{ V}$ (op-amp + input to GND). $R_f = 500 \Omega$. Worst case uses $R_DAC = 689 \Omega$ ($774 \Omega - 11\%$). Output swing limit $\approx \pm 2.7 \text{ V}$ (within 600 mV of $\pm 3.3 \text{ V}$ rails).

Quantity	Nominal ($R_DAC = 774 \Omega$)	Worst case ($R_DAC = 689 \Omega$)
DC current per pin, $1.65 \text{ V} / R_DAC$	2.13 mA	2.39 mA
Audio peak current per pin	1.93 mA	2.17 mA
Max R_f , $2.7 \text{ V} / (I_DC + I_AC, pk)$	665 Ω	592 Ω

With $V_G = 0 \text{ V}$, the DC current flows through R_f and shifts the I/V output's idle point below 0 V. It is common to both legs of a channel and cancels in the differential summing stage.

Feedback capacitor: $C_f = 1.5 \text{ nF}$, C0G ceramic, in parallel with R_f . Target $\leq 0.05 \text{ dB}$ loss at 20 kHz from the I/V stage alone.

Quantity	Value
I/V corner $f_c = 1 / (2\pi R_f C_f)$	$\approx 212 \text{ kHz}$
Loss at 20 kHz, $10 \log_{10}(1 + (20 \text{ kHz} / f_c)^2)$	$\approx 0.038 \text{ dB}$

Equations

$$V_{out} = -I \cdot R_f \cdot \frac{1}{2\pi R_f C_f} \cdot \frac{1}{\log_{10}\left(\frac{V_1}{V_2}\right)} \cdot \text{dB}$$

Open Problems

- ☐ **Linearity at ±3.3 V.** Confirm THD on the prototype.

Rev Log

Rev	Date	Change
7	2026-09-24	C_f = 1.5 nF C0G selected (f_c ≈ 212 kHz, ≈ 0.038 dB loss at 20 kHz). C_f open problem closed.
6	2026-09-24	V_G = 0 V and R_f = 500 Ω selected. I/V Operating Point section added. DC offset and R_f open problems closed.
5	2026-09-24	OPA1692, OPA4134, and SSM2135 source links removed.
4	2026-09-24	LTspice model no longer required: constraint and model-check open problem removed.
3	2026-09-24	Op-amp recorded in Part Identification: OPA1612AIDRGR (8-SON). Package open problem closed.
2	2026-09-24	Op-amp candidates section removed.
1	2026-09-24	OPA1612 selected for the I/V stage. OPA1692 kept as evaluated alternative; OPA4134 rejected. Open problems updated.
0	2026-09-24	Initial release. Signal chain, DAC output data, candidate evaluation, open problems.

Sources

- ES9038Q2M Datasheet v1.4 (ESS), p. 52 (project file)
- [OPA1612 product page, TI](#)